A Closed-Form Solution to Asymmetric Motion Profile Allowing Acceleration Manipulation

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ABSTRACT

This research provides a closed-form analytical solution for asymmetric S-curve motion profiles. Traditional S-curves assume symmetric acceleration/deceleration, but real robotic joints often experience different gravitational forces depending on direction, requiring asymmetric profiling.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides advanced mathematical formulas to implement gravity-compensated asymmetric S-curve trajectories on the ESP32.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by K. H. Rew provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.